

Experiment 'p_4_8_seperate_lines' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_8_seperate_lines

Natural language statement: null

Method signature: p_4_8_seperate_lines(s: string) returns (chars: string)

Ensures

- `|chars| == 2 * |s|`
- `forall i :: 0 <= i < |s| ==> i % 2 == 1 ==> chars[i] == '\n'`
- `forall i :: 0 <= i < |s| ==> i % 2 == 0 ==> chars[i] == s[i / 2]`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_4_8_seperate_lines(s: string) returns (chars: string)`

The method should respect the following contract:

`ensures |chars| == 2 * |s|, ensures forall i :: 0 <= i < |s| ==> i % 2 == 1 ==> chars[i]`
 ↪ `== '\n', ensures forall i :: 0 <= i < |s| ==> i % 2 == 0 ==> chars[i] == s[i / 2]`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnMFRk4H1ZDL DY637i98uUCbqmKl7

Model: gpt-5.5-2026-04-23

Created at: 1780655965

Finish reason: stop

Usage: 374 tokens in, and 1099 tokens out

```
{
  chars := seq(2 * |s|, i requires 0 <= i < 2 * |s| =>
    if i % 2 == 0 then s[i / 2] else '\n');
}
```

Final Program

```
method p_4_8_seperate_lines(s: string) returns (chars: string)
  ensures |chars| == 2 * |s|
  ensures forall i :: 0 <= i < |s| ==> i % 2 == 1 ==> chars[i] == '\n'
  ensures forall i :: 0 <= i < |s| ==> i % 2 == 0 ==> chars[i] == s[i / 2]
{
  chars := seq(2 * |s|, i requires 0 <= i < 2 * |s| =>
    if i % 2 == 0 then s[i / 2] else '\n');
}
```

Total Token Usage

Input tokens: 374

Output tokens: 1099

Reasoning tokens: 1024

Sum of ‘total tokens’: 1473

Experiment Timings

Overall Experiment started at 1780655965283, ended at 1780656004710, lasting 39427ms (39.43 seconds)

Iteration #1 started at 1780655965286, ended at 1780656004710, lasting 39424ms (39.42 seconds)